

DOLCE+DnS Ultralite

URI

<http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Version Information

3.34

Ontology RDF

[RDF \(turtle\)](#)

Description

The DOLCE+DnS Ultralite ontology. It is a simplification of some parts of the DOLCE Lite-Plus library (cf. <http://www.ontologydesignpatterns.org/ont/dul/DLP397.owl>). Main aspects in which DOLCE+DnS Ultralite departs from DOLCE Lite-Plus are the following:

- The names of classes and relations have been made more intuitive
- The DnS-related part is closer to the newer 'constructive DnS' ontology (<http://www.ontologydesignpatterns.org/ont/dul/cDnS.owl>).
- Temporal and spatial relations are simplified
- Qualities and regions are more relaxed than in DOLCE-Full: they can be used as attributes of any entity; an axiom states that each quality has a region
- Axiomatization makes use of simpler constructs than DOLCE Lite-Plus
- The architecture of the ontology is pattern-based, which means that DOLCE+DnS Ultralite is also available in modules, called 'content ontology design patterns', which can be applied independently in the design of domain ontologies (cf. <http://www.ontologydesignpatterns.org>). If many modules are needed in a same ontology project, it is anyway useful to use this integrated version.

The final result is a lightweight, easy-to-apply foundational ontology for modeling either physical or social contexts. Several extensions of DOLCE+DnS Ultralite have been designed:

- Information objects: <http://www.ontologydesignpatterns.org/ont/dul/IOLite.owl>
- Systems: <http://www.ontologydesignpatterns.org/ont/dul/SystemsLite.owl>
- Plans: <http://www.ontologydesignpatterns.org/ont/dul/PlansLite.owl>
- Legal domain: <http://www.ontologydesignpatterns.org/ont/dul/CLO/CoreLegal.owl>
- Lexical and semiotic domains: http://www.ontologydesignpatterns.org/ont/lmm/LMM_L2.owl
- DOLCE-Zero: <http://www.ontologydesignpatterns.org/ont/d0.owl>

<http://www.ontologydesignpatterns.org/ont/d0.owl> is a commonsense-oriented generalisation of some top-level classes, which allows to use DOLCE with tolerance against ambiguities like abstract vs. concrete information, locations vs. physical artifacts, event occurrences vs. event types, events vs. situations, qualities vs. regions, etc.; etc.

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Overview

Pictures say 1,000 words

Figure 1: Ontology overview

Classes

↑

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Abstract ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Abstract	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Any Entity that cannot be located in space-time. E.g. mathematical entities: formal semantics elements, regions within dimensional spaces, etc.	
Super-classes	Entity ^c	
Sub-classes	Formal entity ^c Region ^c	

Action ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Action	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	An Event with at least one Agent that isParticipantIn it, and that executes a Task that typically isDefinedIn a Plan, Workflow, Project, etc.	
Super-classes	Event ^c	
Restrictions	has participant ^{op} some Agent ^c executes task ^{op} min 1	
In domain of	is action included in ^{op} executes task ^{op}	
In range of	includes action ^{op} is executed in ^{op}	

Agent ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Agent	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Additional comment: a computational agent can be considered as a PhysicalAgent that realizes a certain class of algorithms (that can be considered as instances of InformationObject) that allow to obtain some behaviors that are considered typical of agents in general. For an ontology of computational objects based on DOLCE see e.g. http://www.loa-cnr.it/COS/COS.owl , and http://www.loa-cnr.it/KCO/KCO.owl .	
Super-classes	Object ^c	
Sub-classes	Person ^c Physical agent ^c Social agent ^c	
In domain of	is agent included in ^{op} is agent involved in ^{op} conceptualizes ^{op} acts for ^{op}	
In range of	acts through ^{op} includes agent ^{op}	

involves agent ^{op} is conceptualized by ^{op}	
Amount ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Amount
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A quantity, independently from how it is measured, computed, etc.
Super-classes	Region ^c
Biological object ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#BiologicalObject
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-classes	Physical body ^c
Sub-classes	Organism ^c
Chemical object ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#ChemicalObject
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-classes	Physical body ^c
Classification ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Classification
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A special kind of Situation that allows to include time indexing for the classifies relation in situations. For example, if a Situation s 'my old cradle is used in these days as a flower pot' isSettingFor the entity 'my old cradle' and the TimeIntervals '8June2007' and '10June2007', and we know that s satisfies a functional Description for aesthetic objects, which defines the Concepts 'flower pot' and 'flower', then we also need to know what concept classifies 'my old cradle' at what time. In order to solve this issue, we need to create a sub-situation s' for the classification time: 'my old cradle is a flower pot in 8June2007'. Such sub-situation s' isPartOf s.
Super-classes	TimeIndexedRelation ^c
Restrictions	is setting for ^{op} some Time interval ^c is setting for ^{op} some Concept ^c is setting for ^{op} some Entity ^c
Collection ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Collection
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl

Description	Any container for entities that share one or more common properties. E.g. "stone objects", "the nurses", "the Louvre Aegyptian collection", all the elections for the Italian President of the Republic. A collection is not a logical class: a collection is a first-order entity, while a class is second-order. A collection is neither an aggregate of its member entities (see e.g. ObjectAggregate class).
Super-classes	Social object ^c
Restrictions	has part ^{op} only Collection ^c
Sub-classes	Type collection ^c Collective ^c Configuration ^c
In domain of	is unified by ^{op} has member ^{op} is covered by ^{op} is characterized by ^{op}
In range of	covers ^{op} unifies ^{op} is member of ^{op} characterizes ^{op}

Collective ^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Collective
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Collection whose members are agents, e.g. "the nurses", "the Italian rockabilly fans". Collectives, facon de parler, can act as agents, although they are not assumed here to be agents (they are even disjoint from the class SocialAgent). This is represented by admitting collectives in the range of the relations having Agent in their domain or range.
Super-classes	Collection ^c
Restrictions	has member ^{op} only Agent ^c

Collective agent ^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#CollectiveAgent
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A SocialAgent that is actedBy agents that are (and act as) members of a Collective. A collective agent can have roles that are also roles of those agents. For example, in sociology, a 'group action' is the situation in which a number of people (that result to be members of a collective) in a given area behave in a coordinated way in order to achieve a (often common) goal. The Agent in such a Situation is not single, but a CollectiveAgent (a Group). This can be generalized to the notion of social movement, which assumes a large Community or even the entire Society as agents. The difference between a CollectiveAgent and an Organization is that a Description that introduces a CollectiveAgent is also one that unifies the corresponding Collective. In practice, this difference makes collective agents 'less stable' than organizations, because they have a dedicated, publicly recognizable Description that is conceived to introduce them.
Super-classes	Social agent ^c
Restrictions	acts through ^{op} some Agent ^c is introduced by ^{op} some Description ^c

Sub-classes	Community ^c Group ^c
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Community ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Community
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-classes	Collective agent ^c

Concept ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Concept is a SocialObject, and isDefinedIn some Description; once defined, a Concept can be used in other Description(s). If a Concept isDefinedIn exactly one Description, see the LocalConcept class. The classifies relation relates Concept(s) to Entity(s) at some TimeInterval
Super-classes	Social object ^c
Restrictions	is defined in ^{op} some Description ^c has part ^{op} only Concept ^c
Sub-classes	Event type ^c Role ^c Local concept ^c Parameter ^c
In domain of	is related to concept ^{op} is concept used in ^{op} is defined in ^{op} has parameter ^{op} is subordinated to ^{op} is superordinated to ^{op} covers ^{op} classifies ^{op} is concept expressed by ^{op} characterizes ^{op}
In range of	defines ^{op} uses concept ^{op} is subordinated to ^{op} is superordinated to ^{op} expresses concept ^{op} is covered by ^{op} is characterized by ^{op} is related to concept ^{op} is classified by ^{op} is parameter for ^{op}

Configuration ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Configuration
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A collection whose members are 'unified', i.e. organized according to a certain schema that can be represented by a Description. Typically, a configuration is the collection that emerges out of a

composed entity: an industrial artifact, a plan, a discourse, etc.

E.g. a physical book has a configuration provided by the part-whole schema that holds together its cover, pages, ink. That schema, based on the individual relations between the book and its parts, can be represented in a reified way by means of a (structural) description, which is said to 'unify' the book configuration.

Super-classes

[Collection](#)^c

Contract^c

[Classes](#)

URI

<http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Contract>

Is Defined By

<http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description

(The content of) an agreement between at least two agents that play a Party Role, about some contract object (a Task to be executed).

Super-classes

[Description](#)^c

Description^c

[Classes](#)

URI

<http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description>

Is Defined By

<http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description

A Description is a SocialObject that represents a conceptualization. It can be thought also as a 'descriptive context' that uses or defines concepts in order to create a view on a 'relational context' (cf. Situation) out of a set of data or observations. For example, a Plan is a Description of some actions to be executed by agents in a certain way, with certain parameters; a Diagnosis is a Description that provides an interpretation for a set of observed entities, etc. Descriptions 'define' or 'use' concepts, and can be 'satisfied' by situations.

Super-classes

[Social object](#)^c

Sub-classes

[Design](#)^c

[Contract](#)^c

[Theory](#)^c

[Narrative](#)^c

[Goal](#)^c

[Norm](#)^c

[Relation](#)^c

[Right](#)^c

[Method](#)^c

[Plan](#)^c

[Diagnosis](#)^c

In domain of

[defines](#)^{op}

[is satisfied by](#)^{op}

[uses concept](#)^{op}

[unifies](#)^{op}

[describes](#)^{op}

[is related to description](#)^{op}

[is expanded in](#)^{op}

[introduces](#)^{op}

[defines task](#)^{op}

[defines role](#)^{op}

[expands](#)^{op}

In range of

[is task defined in](#)^{op}

[is related to description](#)^{op}

[is role defined in](#)^{op}

[is expanded in](#)^{op}

[is unified by](#)^{op}
[is concept used in](#)^{op}
[expands](#)^{op}
[satisfies](#)^{op}
[is described by](#)^{op}
[is defined in](#)^{op}
[is introduced by](#)^{op}

Design^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Design
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Description of the Situation, in terms of structure and function, held by an Entity for some reason. A design is usually accompanied by the rationales behind the construction of the designed Entity (i.e. of the reasons why a design is claimed to be as such). For example, the actual design (a Situation) of a car or of a law is based on both the specification (a Description) of the structure, and the rationales used to construct cars or laws. While designs typically describe entities to be constructed, they can also be used to describe 'refunctionalized' entities, or to hypothesize unknown functions. For example, a cradle can be refunctionalized as a flowerpot based on a certain home design.
Super-classes	Description ^c

Designed artifact^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#DesignedArtifact
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A PhysicalArtifact that is also described by a Design. This excludes simple recycling or refunctionalization of natural objects. Most common sense 'artifacts' can be included in this class: cars, lamps, houses, chips, etc.
Super-classes	Physical artifact ^c
Restrictions	is described by ^{op} some Design ^c
Sub-classes	DesignedSubstance ^c

DesignedSubstance^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#DesignedSubstance
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-classes	Functional substance ^c Designed artifact ^c

Diagnosis^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Diagnosis
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Description of the Situation of a system, usually applied in order to control a normal behaviour, or to explain a notable behavior (e.g. a functional breakdown).

Super-classes	Description ^c
Entity ^c	
# Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Anything: real, possible, or imaginary, which some modeller wants to talk about for some purpose.
Sub-classes	Event ^c Object ^c Situation ^c Quality ^c InformationEntity ^c Abstract ^c
In domain of	has part ^{op} is classified by ^{op} associatedWith ^{op} has location ^{op} is described by ^{op} directly follows ^{op} is observable at ^{op} overlaps ^{op} has data value ^{dp} has component ^{op} has constituent ^{op} is reference of information realized by ^{op} has constraint ^{op} near to ^{op} precedes ^{op} is in the same setting as ^{op} is reference of ^{op} far from ^{op} is member of ^{op} has quality ^{op} is part of ^{op} has region ^{op} follows ^{op} is location of ^{op} has common boundary ^{op} directly precedes ^{op} is component of ^{op} has setting ^{op} is constituent of ^{op}
In range of	is setting for ^{op} describes ^{op} overlaps ^{op} has component ^{op} is location of ^{op} is about ^{op} directly precedes ^{op} classifies ^{op} has constituent ^{op} near to ^{op} precedes ^{op} is region for ^{op} is in the same setting as ^{op} realizes information about ^{op} is constraint for ^{op} far from ^{op}

[is part of](#)^{op}
[has member](#)^{op}
[follows](#)^{op}
[is time of observation of](#)^{op}
[has common boundary](#)^{op}
[is component of](#)^{op}
[is quality of](#)^{op}
[is constituent of](#)^{op}
[has part](#)^{op}
[associatedWith](#)^{op}
[has location](#)^{op}
[directly follows](#)^{op}

Event^c# [Classes](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description

Any physical, social, or mental process, event, or state.

More theoretically, events can be classified in different ways, possibly based on 'aspect' (e.g. stative, continuous, accomplishment, achievement, etc.), on 'agentivity' (e.g. intentional, natural, etc.), or on 'typical participants' (e.g. human, physical, abstract, food, etc.). Here no special direction is taken, and the following explains why: events are related to observable situations, and they can have different views at a same time. If a position has to be suggested here anyway, the participant-based classification of events seems the most stable and appropriate for many modelling problems.

(1) Alternative aspectual views

Consider a same event 'rock erosion in the Sinni valley': it can be conceptualized as an accomplishment (what has brought a certain state to occur), as an achievement (the state resulting from a previous accomplishment), as a punctual event (if we collapse the time interval of the erosion into a time point), or as a transition (something that has changed from a state to a different one). In the erosion case, we could therefore have good motivations to shift from one aspect to another: a) causation focus, b) effectual focus, c) historical condensation, d) transition (causality).

The different views refer to the same event, but are still different: how to live with this seeming paradox? A typical solution e.g. in linguistics (cf. Levin's aspectual classes) and in DOLCE Full (cf. WonderWeb D18 axiomatization) is to classify events based on aspectual differences. But this solution would create different identities for a same event, where the difference is only based on the modeller's attitude. An alternative solution is suggested here, and exploits the notion of (observable) Situation; a Situation is a view, consistent with a Description, which can be observed of a set of entities. It can also be seen as a 'relational context' created by an observer on the basis of a 'frame'. Therefore, a Situation allows to create a context where each particular view can have a proper identity, while the Event preserves its own identity. For example, ErosionAsAccomplishment is a Situation where rock erosion is observed as a process leading to a certain achievement: the conditions (roles, parameters) that suggest such view are stated in a Description, which acts as a 'theory of accomplishments'. Similarly, ErosionAsTransition is a Situation where rock erosion is observed as an event that has changed a state to another: the conditions for such interpretation are stated in a different Description, which acts as a 'theory of state transitions'. Consider that in no case the actual event is changed or enriched in parts by the aspectual view.

(2) Alternative intentionality views

Similarly to aspectual views, several intentionality views can be provided for a same Event. For example, one can investigate if an avalanche has been caused by immediate natural forces, or if there is any hint of an intentional effort to activate those natural forces. Also in this case, the Event as such has no different identities, while the causal analysis generates situations with different identities, according to what Description is taken for interpreting the Event. On the other hand, if the possible actions of an Agent causing the starting of an avalanche are taken as parts of the Event, then this makes its identity change, because we are adding a part to it. Therefore, if intentionality is a criterion to classify events or not, this depends on if an ontology designer wants to consider causality as a relevant dimension for events' identity.

(3) Alternative participant views

A slightly different case is when we consider the basic participants to an Event. In this case, the identity of the Event is affected by the participating objects, because it depends on them. For example, if snow, mountain slopes, wind, waves, etc. are considered as an avalanche basic participants, or if we also want to add water, human agents, etc., that makes the identity of an avalanche change. Anyway, this approach to event classification is based on the designer's choices, and more accurately mirrors lexical or commonsense classifications (see. e.g. WordNet 'supersenses' for verb synsets).

Ultimately, this discussion has no end, because realists will keep defending the idea that events in reality are not changed by the way we describe them, while constructivists will keep defending the idea that, whatever 'true reality' is about, it can't be modelled without the theoretical burden of how we observe and describe it. Both positions are in principle valid, but, if taken too radically, they focus on issues that are only partly relevant to the aim of computational ontologies, which assist domain experts in representing a certain portion of reality according to their own assumptions and requirements.

For this reason, in this ontology version of DOLCE, both events and situations are allowed, together with descriptions (the reason for the inclusion of the D&S framework in DOLCE), in order to encode the modelling needs, independently from the position (if any) chosen by the model designer.

Super-classes	Entity ^c
Restrictions	has constituent ^{op} <i>only</i> Event ^c has part ^{op} <i>only</i> Event ^c has participant ^{op} <i>some</i> Object ^c has time interval ^{op} <i>some</i> Time interval ^c
Sub-classes	Action ^c Process ^c
In domain of	has event date ^{dp} is event included in ^{op} has time interval ^{op} has participant ^{op} involves agent ^{op}
In range of	is agent involved in ^{op} is time interval of ^{op} is participant in ^{op} includes event ^{op}

Event type^c

[# Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#EventType
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Concept that classifies an Event . An event type describes how an Event should be interpreted, executed, expected, seen, etc., according to the Description that the EventType isDefinedIn (or used in)
Super-classes	Concept ^c
Restrictions	classifies ^{op} <i>only</i> Event ^c
Sub-classes	Task ^c

Formal entity^c

[# Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#FormalEntity
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Entities that are formally defined and are considered independent from the social context in which they are used. They cannot be localized in space or time. Also called 'Platonic entities'. Mathematical and logical entities are included in this class: sets, categories, tuples, constants, variables, etc. Abstract formal entities are distinguished from information objects, which are supposed to be part of a social context, and are localized in space and time, therefore being (social) objects. For example, the class 'Quark' is an abstract formal entity from the purely set-theoretical perspective, but it is an InformationObject from the viewpoint of ontology design, when e.g. implemented in a logical language like OWL. Abstract formal entities are also distinguished from Concept(s), Collection(s), and Description(s), which are part of a social context, therefore being SocialObject(s) as well. For example, the class 'Quark' is an abstract FormalEntity from the

purely set-theoretical perspective, but it is a Concept within history of science and cultural dynamics.

These distinctions allow to represent two different notions of 'semantics': the first one is abstract and formal ('formal semantics'), and formally interprets symbols that are about entities whatsoever; for example, the term 'Quark' is About the Collection of all quarks, and that Collection is FormalGroundingFor the abstract class 'Quark' (in the extensional sense). The second notion is social, localized in space-time ('social semantics'), and can be used to interpret entities in the intensional sense. For example, the Collection of all quarks is CoveredBy the Concept 'Quark', which is also expressed by the term 'Quark'.

Super-classes [Abstract](#)^c

Sub-classes [Set](#)^c

Functional substance^c

[# Classes](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#FunctionalSubstance>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Super-classes [Substance](#)^c

Sub-classes [DesignedSubstance](#)^c

Goal^c

[# Classes](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Goal>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
The Description of a Situation that is desired by an Agent, and usually associated to a Plan that describes how to actually achieve it

Super-classes [Description](#)^c

Group^c

[# Classes](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Group>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
A CollectiveAgent whose acting agents conceptualize a same SocialRelation .

Super-classes [Collective agent](#)^c

Restrictions [is described by](#)^{op} [some Plan](#)^c

InformationEntity^c

[# Classes](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationEntity>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description	A piece of information, be it concretely realized or not. It is a catchall class, intended to bypass the ambiguities of many data or text that could denote either a an expression or a concrete realization of that expression. In a semiotic model, there is no special reason to distinguish between them, however we may want to distinguish between a pure information content (e.g. the 3rd Gymnopedie by Satie), and its possible concrete realizations as a music sheet, a piano execution, the reproduction of the execution, its publishing as a record, etc.).
Super-classes	Entity ^c
Sub-classes	Information realization ^c Information object ^c

Information object ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationObject
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A piece of information, such as a musical composition, a text, a word, a picture, independently from how it is concretely realized.
Super-classes	Social object ^c InformationEntity ^c
In domain of	is about ^{op} expresses concept ^{op} is realized by ^{op} expresses ^{op}
In range of	is concept expressed by ^{op} realizes ^{op} is reference of ^{op} is expressed by ^{op}

Information realization ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationRealization
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A concrete realization of an InformationObject, e.g. the written document (object) containing the text of a law, a poetry reading (event), the dark timbre (quality) of a sound (event) in the execution (event) of a musical composition, realizing a 'misterioso' tempo indication. The realization of an information object also realizes information about itself. This is a special semiotic feature, which allows to avoid a traditonal paradox, by which an information is often supposed to be about itself besides other entities (e.g. the information object 'carpe diem' is about its meaning in Horace's Odes (let alone its fortune in Western culture and beyond), but also about its expression in context: 'dum loquimur, fugerit invida aetas: carpe diem, quam minimum credula postero', with the sound and emotional relations that it could activate. This is expressed in OWL2 with a local reflexivity axiom of the dul:InformationRealization class.
Super-classes	InformationEntity ^c (Event ^c or Physical object ^c or Quality ^c)
Restrictions	realizes ^{op} <i>some</i> Information object ^c realizesSelfInformation ^{op}
In domain of	realizes ^{op} realizes information about ^{op} concretely expresses ^{op}

In range of is concretely expressed by ^{op} is reference of information realized by ^{op} is realized by ^{op}	
Local concept ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#LocalConcept
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Concept that isDefinedIn exactly 1 Description. For example, the Concept 'coffee' in a 'preparesCoffee' relation can be defined in that relation, and for all other Description(s) that use it, the isConceptUsedIn property should be applied. Notice therefore that not necessarily all Concept(s) isDefinedIn exactly 1 Description.
Super-classes	Concept ^c
Method ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Method
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A method is a Description that defines or uses concepts in order to guide carrying out actions aimed at a solution with respect to a problem. It is different from a Plan, because plans could be carried out in order to follow a method, but a method can be followed by executing alternative plans.
Super-classes	Description ^c
Narrative ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Narrative
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-classes	Description ^c
Natural person ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#NaturalPerson
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A person in the physical commonsense intuition: 'have you seen that person walking down the street?'
Super-classes	Person ^c Physical agent ^c
Norm ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Norm
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl

Description	A social norm.
Super-classes	Description ^c

Object ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Object
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any physical, social, or mental object, or a substance. Following DOLCE Full, objects are always participating in some event (at least their own life), and are spatially located.
Super-classes	Entity ^c
Restrictions	has location ^{op} some Entity ^c has constituent ^{op} only Object ^c is classified by ^{op} only Role ^c has part ^{op} only Object ^c is participant in ^{op} some Event ^c
Sub-classes	Social object ^c Physical object ^c Agent ^c
In domain of	co-participates with ^{op} is participant in ^{op} is object included in ^{op} has role ^{op}
In range of	co-participates with ^{op} has participant ^{op} is role of ^{op} includes object ^{op}

ObjectAggregate ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#ObjectAggregate
Description	An aggregate of distributed objects, members of a same Collection, e.g. the stars in a constellation, the parts of a car, the employees of a company, the entries from an encyclopedia, the concepts expressed in a speech, etc. It cannot be defined by means of an equivalence axiom, because it'd require the same Collection for all members, an axiom that cannot be expressed in OWL.
Super-classes	(Object ^c and ub3443bL499C13 ^c)

Organism ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Organism
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A physical objects with biological characteristics, typically that organisms can self-reproduce.
Super-classes	Physical agent ^c Biological object ^c

Organization ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Organization	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	An internally structured, conventionally created SocialAgent, needing a specific Role and Agent that plays it, in order to act.	
Super-classes	Social agent ^c	

Parameter ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Parameter	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A Concept that classifies a Region; the difference between a Region and a Parameter is that regions represent sets of observable values, e.g. the height of a given building, while parameters represent constraints or selections on observable values, e.g. 'VeryHigh'. Therefore, parameters can also be used to constrain regions, e.g. VeryHigh on a subset of values of the Region Height applied to buildings, or to add an external selection criterion , such as measurement units, to regions, e.g. Meter on a subset of values from the Region Length applied to the Region Length applied to roads.	
Super-classes	Concept ^c	
Restrictions	classifies ^{op} only Region ^c has part ^{op} only Parameter ^c	
Sub-classes	Unit of measure ^c	
In domain of	is constraint for ^{op} has parameter data value ^{dp} is parameter for ^{op} parametrizes ^{op}	
In range of	is parametrized by ^{op} has parameter ^{op} has constraint ^{op}	

parthood ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Parthood	
Description	A special kind of Situation that allows to include time indexing for the hasPart relation in situations. For example, if a Situation s 'finally, my bike has a luggage rack' isSettingFor the entity 'my bike' and the TimeIntervals 'now', or more specifically '29March2021', we need to have a time-index the part relation. With Parthood, we use includesWhole and includesPart properties. This can be done similarly for other arguments of parthood, e.g. location, configuration, topology, etc. Concerning the possible property characteristics reused from mereology (transitivity, asymmetry, reflexivity), they need to be implemented by means of rules (or, in a limited way, property chains using the binary hasPart or hasProperPart properties). A key is also added to ensure identification constraints of time-indexed parthood.	
Super-classes	TimeIndexedRelation ^c	
Restrictions	includesWhole ^{op} some Entity ^c includesPart ^{op} some Entity ^c	

Pattern ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Pattern	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Any invariance detected from a dataset, or from observation; also, any invariance proposed based on top-down considerations. E.g. patterns detected and abstracted by an organism, by pattern recognition algorithms, by machine learning techniques, etc. An occurrence of a pattern is an 'observable', or detected Situation	
Super-classes	Relation ^c	

Person ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Person	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Persons in commonsense intuition, which does not apparently distinguish between either natural or social persons.	
Super-classes	Agent ^c	
Sub-classes	Social person ^c Natural person ^c	

Personification ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Personification	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A social entity with agentive features, but whose status is the result of a cultural transformation from e.g. a PhysicalObject, an Event, an Abstract, another SocialObject, etc. For example: the holy grail, deus ex machina, gods, magic wands, etc.	
Super-classes	Social agent ^c	

Physical agent ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#PhysicalAgent	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A PhysicalObject that is capable of self-representing (conceptualizing) a Description in order to plan an Action. A PhysicalAgent is a substrate for (actsFor) a Social Agent	
Super-classes	Agent ^c Physical object ^c	
Sub-classes	Organism ^c Natural person ^c	

Physical artifact ^c		# Classes

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#PhysicalArtifact
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any PhysicalObject that isDescribedBy a Plan . This axiomatization is weak, but allows to talk of artifacts in a very general sense, i.e. including recycled objects, objects with an intentional functional change, natural objects that are given a certain function, even though they are not modified or structurally designed, etc. PhysicalArtifact(s) are not considered disjoint from PhysicalBody(s), in order to allow a dual classification when needed. E.g., FunctionalSubstance(s) are included here as well. Immaterial (non-physical) artifacts (e.g. texts, ideas, cultural movements, corporations, communities, etc. can be modelled as social objects (see SocialObject), which are all 'artifactual' in the weak sense assumed here.
Super-classes	Physical object ^c
Restrictions	is described by ^{op} some Plan ^c
Sub-classes	Designed artifact ^c

Physical attribute^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#PhysicalAttribute
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Physical value of a physical object, e.g. density, color, etc.
Super-classes	Region ^c
Restrictions	is region for ^{op} only Physical object ^c

Physical body^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#PhysicalBody
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Physical bodies are PhysicalObject(s), for which we tend to neutralize any possible artifactual character. They can have several granularity levels: geological, chemical, physical, biological, etc.
Super-classes	Physical object ^c
Sub-classes	Chemical object ^c Substance ^c Biological object ^c

Physical object^c# [Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#PhysicalObject
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any Object that has a proper space region. The prototypical physical object has also an associated mass, but the nature of its mass can greatly vary based on the epistemological status of the object (scientifically measured, subjectively possible, imaginary).
Super-classes	Object ^c

Restrictions	has part ^{op} <i>only</i> Physical object ^c
Sub-classes	Physical artifact ^c Physical agent ^c Physical body ^c Physical place ^c

Physical place ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#PhysicalPlace
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A physical object that is inherently located; for example, a water area.
Super-classes	Physical object ^c

Place ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Place
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Socially or cognitively dependent locations: political geographic entities (Rome, Lesotho), and non-material locations determined by the presence of other entities ("the area close to Rome") or of pivot events or signs ("the area where the helicopter fell"), as well as identified as complements to other entities ("the area under the table"), etc. In this generic sense, a Place is a 'dependent' location. For 'non-dependent' locations, cf. the PhysicalPlace class. For an abstract (dimensional) location, cf. the SpaceRegion class.
Super-classes	Social object ^c
Restrictions	is location of ^{op} <i>min</i> 1

Plan ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Plan
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Description having an explicit Goal, to be achieved by executing the plan
Super-classes	Description ^c
Restrictions	has component ^{op} <i>some</i> Goal ^c
Sub-classes	Workflow ^c Project ^c

Plan execution ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#PlanExecution
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl

Description	Plan executions are situations that proactively satisfy a plan. Subplan executions are proper parts of the whole plan execution.
Super-classes	Situation ^c

Process ^c	# Classes
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Process
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	This is a placeholder for events that are considered in their evolution, or anyway not strictly dependent on agents, tasks, and plans. See Event class for some thoughts on classifying events. See also 'Transition'.
Super-classes	Event ^c

Project ^c	# Classes
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Project
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Plan that defines Role(s), Task(s), and a specific structure for tasks to be executed in relation to goals to be achieved, in order to achieve the main goal of the project. In other words, a project is a plan with a subgoal structure and multiple roles and tasks.
Super-classes	Plan ^c
Restrictions	defines role ^{op} some Role ^c defines task ^{op} some Task ^c

Quality ^c	# Classes
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Quality
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any aspect of an Entity (but not a part of it), which cannot exist without that Entity. For example, the way the surface of a specific PhysicalObject looks like, or the specific light of a place at a certain time, are examples of Quality, while the encoding of a Quality into e.g. a PhysicalAttribute should be modeled as a Region. From the design viewpoint, the Quality-Region distinction is useful only when individual aspects of an Entity are considered in a domain of discourse. For example, in an automotive context, it would be irrelevant to consider the aspects of car windows for a specific car, unless the factory wants to check a specific window against design parameters (anomaly detection). On the other hand, in an antiques context, the individual aspects for a specific piece of furniture are a major focus of attention, and may constitute the actual added value, because the design parameters for old furniture are often not fixed, and may not be viewed as 'anomalies'.
Super-classes	Entity ^c
Restrictions	has part ^{op} only Quality ^c has region ^{op} some Region ^c is quality of ^{op} some Entity ^c has constituent ^{op} only Quality ^c
In domain of	is quality of ^{op}
In range of	has quality ^{op}

Region ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Region	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Any region in a dimensional space (a dimensional space is a maximal Region), which can be used as a value for a quality of an Entity . For example, TimeInterval, SpaceRegion, PhysicalAttribute, Amount, SocialAttribute are all subclasses of Region. Regions are not data values in the ordinary knowledge representation sense; in order to get patterns for modelling data, see the properties: representsDataValue and hasDataValue	
Super-classes	Abstract ^c	
Restrictions	overlaps ^{op} only Region ^c precedes ^{op} only Region ^c has constituent ^{op} only Region ^c has part ^{op} only Region ^c	
Sub-classes	Social attribute ^c Amount ^c Space region ^c SpatioTemporalRegion ^c Physical attribute ^c Time interval ^c	
In domain of	is region for ^{op} is parametrized by ^{op} has region data value ^{dp}	
In range of	has region ^{op} parametrizes ^{op}	

Relation ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Relation	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Relations are descriptions that can be considered as the counterpart of formal relations (that are included in the FormalEntity class). For example, 'givingGrantToInstitution(x,y,z)' with three argument types: Provider(x),Grant(y),Recipient(z), can have a Relation counterpart: 'GivingGrantToInstitution', which defines three Concept instances: Provider,Grant,Recipient. Since social objects are not formal entities, Relation includes here any 'relation-like' entity in common sense, including social relations.	
Super-classes	Description ^c	
Sub-classes	Social relation ^c Pattern ^c	

Right ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Right	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A legal position by which an Agent is entitled to obtain something from another Agent , under specified circumstances, through an enforcement explicited either in a Law, Contract , etc.	

Super-classes	Description ^c
Restrictions	defines role ^{op} <i>min</i> 2 defines task ^{op} <i>min</i> 1

Role ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Role
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Concept that classifies an Object
Super-classes	Concept ^c
Restrictions	has part ^{op} <i>only</i> Role ^c classifies ^{op} <i>only</i> Object ^c
In domain of	is role defined in ^{op} has task ^{op} is role of ^{op}
In range of	is task of ^{op} defines role ^{op} has role ^{op}

Set ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Set
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-classes	Formal entity ^c

Situation ^c	# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A view, consistent with ('satisfying') a Description, on a set of entities. It can also be seen as a 'relational context' created by an observer on the basis of a 'frame' (i.e. a Description). For example, a PlanExecution is a context including some actions executed by agents according to certain parameters and expected tasks to be achieved from a Plan; a DiagnosedSituation is a context of observed entities that is interpreted on the basis of a Diagnosis, etc. Situation is also able to represent reified n-ary relations, where isSettingFor is the top-level relation for all binary projections of the n-ary relation. If used in a transformation pattern for n-ary relations, the designer should take care of adding (some or all) OWL2 keys, corresponding to binary projections of the n-ary, to a subclass of Situation. Otherwise the 'identification constraint' (Calvanese et al., IJCAI 2001) might be violated.
Super-classes	Entity ^c
Restrictions	satisfies ^{op} <i>some</i> Description ^c
Sub-classes	TimeIndexedRelation ^c Plan execution ^c Transition ^c Workflow execution ^c

In domain of	is setting for ^{op} includes object ^{op} includes event ^{op} includes time ^{op} includes action ^{op} satisfies ^{op} includes agent ^{op}
In range of	is event included in ^{op} is time included in ^{op} is object included in ^{op} is satisfied by ^{op} is action included in ^{op} has setting ^{op} is agent included in ^{op}

Social agent ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialAgent	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Any individual whose existence is granted simply by its social communicability and capability of action (through some PhysicalAgent).	
Super-classes	Agent ^c Social object ^c	
Restrictions	acts through ^{op} <i>some</i> Physical agent ^c	
Sub-classes	Personification ^c Social person ^c Organization ^c Collective agent ^c	
In domain of	acts through ^{op} is introduced by ^{op}	
In range of	acts for ^{op} introduces ^{op}	

Social object ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialObject	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Any Object that exists only within some communication Event, in which at least one PhysicalObject participates in. In other words, all objects that have been or are created in the process of social communication: for the sake of communication (InformationObject), for incorporating new individuals (SocialAgent, Place), for contextualizing or interpreting existing entities (Description, Concept), or for collecting existing entities (Collection). Being dependent on communication, all social objects need to be expressed by some information object (information objects are self-expressing).	
Super-classes	Object ^c	
Restrictions	has part ^{op} <i>only</i> Social object ^c is expressed by ^{op} <i>some</i> Information object ^c	
Sub-classes	Information object ^c Collection ^c	

	Place ^c Social agent ^c Concept ^c Description ^c
In domain of	specializes ^{op} is expressed by ^{op} is specialized by ^{op} is conceptualized by ^{op} is concretely expressed by ^{op}
In range of	conceptualizes ^{op} expresses ^{op} specializes ^{op} is specialized by ^{op} concretely expresses ^{op}

Social attribute ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialObjectAttribute	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Any Region in a dimensional space that is used to represent some characteristic of a SocialObject, e.g. judgment values, social scalars, statistical attributes over a collection of entities, etc.	
Super-classes	Region ^c	
Restrictions	is region for ^{op} <i>only</i> Social object ^c	

Social person ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialPerson	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A SocialAgent that needs the existence of a specific NaturalPerson in order to act (but the lifetime of the NaturalPerson has only to overlap that of the SocialPerson).	
Super-classes	Person ^c Social agent ^c	
Restrictions	acts through ^{op} <i>exactly</i> 1	

Social relation ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialRelation	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	Any social relationship	
Super-classes	Relation ^c	

Space region ^c		# Classes
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SpaceRegion	

Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any Region in a dimensional space that is used to localize an Entity ; i.e., it is not used to represent some characteristic (e.g. it excludes time intervals, colors, size values, judgment values, etc.). Differently from a Place , a space region has a specific dimensional space.
Super-classes	Region ^c

SpatioTemporalRegion ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SpatioTemporalRegion
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-classes	Region ^c
Restrictions	has constituent ^{op} some Time interval ^c has constituent ^{op} some Space region ^c

Substance ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Substance
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any PhysicalBody that has not necessarily specified (designed) boundaries, e.g. a pile of trash, some sand, etc. In this sense, an artistic object made of trash or a dose of medicine in the form of a pill would be a FunctionalSubstance, and a DesignedArtifact, since its boundaries are specified by a Design; aleatoric objects that are outcomes of an artistic process might be still considered DesignedArtifact(s), and Substance(s).
Super-classes	Physical body ^c
Sub-classes	Functional substance ^c

Task ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Task
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	An EventType that classifies an Action to be executed. For example, reaching a destination is a task that can be executed by performing certain actions, e.g. driving a car, buying a train ticket, etc. The actions to execute a task can also be organized according to a Plan that is not the same as the one that defines the task (if any). For example, reaching a destination could be defined by a plan to get on holidays, while the plan to execute the task can consist of putting some travels into a sequence.
Super-classes	Event type ^c
Restrictions	has part ^{op} only Task ^c is task defined in ^{op} only Description ^c is executed in ^{op} only Action ^c is task of ^{op} only Role ^c
In domain of	is executed in ^{op} is task of ^{op} is task defined in ^{op}
In range of	defines task ^{op} has task ^{op}

executes task ^{op}	
Theory ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Theory
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Theory is a Description that represents a set of assumptions for describing something, usually general. Scientific, philosophical, and commonsense theories can be included here. This class can also be used to act as 'naturalized reifications' of logical theories (of course, they will be necessarily incomplete in this case, because second-order entities are represented as first-order ones).
Super-classes	Description ^c
Restrictions	has component ^{op} some Relation ^c
TimeIndexedRelation ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#TimeIndexedRelation
Description	A Situation that includes a time indexing in its setting, so allowing to order any binary relation (property) with time.
Super-classes	Situation ^c
Sub-classes	Classification ^c parthood ^c
Time interval ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#TimeInterval
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any Region in a dimensional space that aims at representing time.
Super-classes	Region ^c
In domain of	is time of observation of ^{op} has interval date ^{dp} is time included in ^{op} is time interval of ^{op}
In range of	is observable at ^{op} has time interval ^{op} includes time ^{op}
Transition ^c # Classes	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Transition
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A transition is a Situation that creates a context for three TimeInterval(s), two additional different Situation(s), one Event, one Process, and at least one Object: the Event is observed as the cause for the transition, one Situation is the state before the transition, the second Situation is the state after the transition, the Process is the invariance under some different transitions (including the one

represented here), in which at least one Object is situated. Finally, the time intervals position the situations and the transitional event in time. This class of situations partly encodes the ontology underlying typical engineering algebras for processes, e.g. Petri Nets. A full representation of the transition ontology is outside the expressivity of OWL, because we would need qualified cardinality restrictions, coreference, property equivalence, and property composition.

Super-classes [Situation](#)^c

Restrictions [is setting for](#)^{op} [some](#) ([Situation](#)^c and [ub3443bL994C17](#)^c)
[includes event](#)^{op} [some](#) [Event](#)^c
[is setting for](#)^{op} [some](#) [Process](#)^c
[is setting for](#)^{op} [min](#) 2 [Situation](#)^c
[includes object](#)^{op} [some](#) [Object](#)^c
[includes time](#)^{op} [min](#) 3 [Time interval](#)^c

Type collection^c

[# Classes](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#TypeCollection>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
 A Collection whose members are the maximal set of individuals that share the same (named) type, e.g. "the gem stones", "the Italians". This class is very useful to apply a variety of the so-called "ClassesAsValues" design pattern, when it is used to talk about the extensional aspect of a class. An alternative variety of the pattern applies to the intensional aspect of a class, and the class Concept should be used instead.

Super-classes [Collection](#)^c

Unit of measure^c

[# Classes](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#UnitOfMeasure>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
 Units of measure are conceptualized here as parameters on regions, which can be valued as datatype values.

Super-classes [Parameter](#)^c

Restrictions [parametrizes](#)^{op} [some](#) [Region](#)^c

Workflow^c

[# Classes](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Workflow>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
 A Plan that defines Role(s), Task(s), and a specific structure for tasks to be executed, usually supporting the work of an Organization

Super-classes [Plan](#)^c

Restrictions [defines task](#)^{op} [some](#) [Task](#)^c
[defines role](#)^{op} [some](#) [Role](#)^c

Workflow execution^c

[# Classes](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#WorkflowExecution
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-classes	Situation ^c

Thing ^c	# Classes
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URI	http://www.w3.org/2002/07/owl#Thing
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Object Properties

↑

[acts for](#) [acts through](#) [associatedWith](#) [characterizes](#) [classifies](#) [conceptualizes](#) [concretely expresses](#) [co-participates with](#) [covers](#) [defines](#) [defines role](#) [defines task](#) [describes](#) [directly follows](#) [directly precedes](#) [executes task](#) [expands](#) [expresses](#) [expresses concept](#) [far from](#) [follows](#) [has common boundary](#) [has component](#) [has constituent](#) [has constraint](#) [has location](#) [has member](#) [has parameter](#) [has part](#) [has participant](#) [has postcondition](#) [has precondition](#) [has proper part](#) [has quality](#) [has region](#) [has role](#) [has setting](#) [has task](#) [has time interval](#) [includes action](#) [includes agent](#) [includes event](#) [includes object](#) [includesPart](#) [includes time](#) [includesWhole](#) [introduces](#) [involves agent](#) [is about](#) [is action included in](#) [is agent included in](#) [is agent involved in](#) [is characterized by](#) [is classified by](#) [is component of](#) [is concept expressed by](#) [is concept used in](#) [is conceptualized by](#) [is concretely expressed by](#) [is constituent of](#) [is constraint for](#) [is covered by](#) [is defined in](#) [is described by](#) [is event included in](#) [is executed in](#) [is expanded in](#) [is expressed by](#) [is introduced by](#) [is location of](#) [is member of](#) [is object included in](#) [is observable at](#) [is parameter for](#) [is parametrized by](#) [is part of](#) [is participant in](#) [is postcondition of](#) [is precondition of](#) [is proper part of](#) [is quality of](#) [is realized by](#) [is reference of](#) [is reference of information realized by](#) [is region for](#) [is related to concept](#) [is related to description](#) [is role defined in](#) [is role of](#) [is satisfied by](#) [is setting for](#) [is specialized by](#) [is subordinated to](#) [is superordinated to](#) [is task defined in](#) [is task of](#) [is time included in](#) [is time interval of](#) [is time of observation of](#) [is unified by](#) [near to](#) [overlaps](#) [parametrizes](#) [precedes](#) [realizes](#) [realizes information about](#) [realizesSelfInformation](#) [is in the same setting as](#) [satisfies](#) [specializes](#) [unifies](#) [uses concept](#)

acts for ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#actsFor
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The relation holding between any Agent, and a SocialAgent. In principle, a SocialAgent requires at least one PhysicalAgent in order to act, but this dependency can be 'delegated'; e.g. a university can be acted for by a department, which on its turn is acted for by physical agents.
Super-properties	associatedWith ^{op}
Inverse properties	acts through ^{op}
Domain(s)	Agent ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialAgent ^c

acts through ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#actsThrough
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The relation holding between a PhysicalAgent and a SocialAgent. In principle, a SocialAgent requires at least one PhysicalAgent in order to act, but this dependency can be 'delegated', e.g. a university can be acted for by a department, which is acted for by physical agents. AKA isActedBy
Super-properties	associatedWith ^{op}
Domain(s)	Social agent ^c

Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Agent ^C
associatedWith ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#associatedWith
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A catch-all object property, useful for alignment and querying purposes. It is declared as both transitive and symmetric, in order to reason an a maximal closure of associations between individuals.
Inverse properties	associatedWith ^{op}
Domain(s)	Entity ^C
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^C
characterizes ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#characterizes
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between concepts and collections, where a Concept is said to characterize a Collection; it corresponds to a link between the (reified) intensional and extensional interpretations of a <i>proper subset</i> of a (reified) class. This is different from covers, because it refers to an interpretation the entire reified class. E.g. the collection of vintage saxophones is characterized by the Concept 'manufactured by hand', while it gets covered by the Concept 'Saxophone' with the Parameter 'Vintage'.
Super-properties	associatedWith ^{op}
Inverse properties	is characterized by ^{op}
Domain(s)	Concept ^C
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Collection ^C
classifies ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#classifies
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a Concept and an Entity, e.g. the Role 'student' classifies a Person 'John'.
Super-properties	associatedWith ^{op}
Inverse properties	is classified by ^{op}
Domain(s)	Concept ^C
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^C
conceptualizes ^{op} # OPs	

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#conceptualizes
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation stating that an Agent is internally representing a SocialObject: situations, descriptions, concepts, etc. E.g., 'John believes in the conspiracy theory'; 'Niels Bohr created the solar-system metaphor for the atomic theory'; 'Jacques assumes all swans are white'; 'the task force members share the attack plan'. Conceptualizations can be distinguished into different forms, primarily based on the type of SocialObject that is conceptualized. Descriptions and concepts can be 'assumed', situations can be 'believed' or 'known', plans can be 'adopted', etc. (see ontology: http://www.ontologydesignpatterns.org/ont/dul/Conceptualization.owl).
Super-properties	associatedWith ^{op}
Inverse properties	is conceptualized by ^{op}
Domain(s)	Agent ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialObject ^c

concretely expresses ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#concretelyExpresses
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an InformationRealization and a Description, e.g. 'the printout of the Italian Constitution concretelyExpresses the Italian Constitution'. It should be supplied also with a rule stating that the InformationRealization realizes an InformationObject that expresses the Description
Super-properties	associatedWith ^{op}
Inverse properties	is concretely expressed by ^{op}
Domain(s)	Information realization ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialObject ^c

co-participates with ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#coparticipatesWith
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between two objects participating in a same Event; e.g., 'Vitas and Jimmy are playing tennis'.
Super-properties	associatedWith ^{op}
Inverse properties	co-participates with ^{op}
Domain(s)	Object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Object ^c

covers ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#covers

Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between concepts and collections, where a Concept is said to cover a Collection; it corresponds to a link between the (reified) intensional and extensional interpretations of a (reified) class. E.g. the collection of vintage saxophones is covered by the Concept 'Saxophone' with the Parameter 'Vintage'.
Super-properties	associatedWith ^{op}
Inverse properties	is covered by ^{op}
Domain(s)	Concept ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Collection ^c

defines ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#defines
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a Description and a Concept, e.g. a Workflow for a governmental Organization defines the Role 'officer', or 'the Italian Traffic Law defines the role Vehicle'.
Super-properties	uses concept ^{op}
Inverse properties	is defined in ^{op}
Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c

defines role ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#definesRole
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a description and a role, e.g. the recipe for a cake defines the role 'ingredient'.
Super-properties	defines ^{op}
Inverse properties	is role defined in ^{op}
Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Role ^c

defines task ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#definesTask
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a description and a task, e.g. the recipe for a cake defines the task 'boil'.
Super-properties	defines ^{op}

Inverse properties	is task defined in ^{op}
Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Task ^c

describes ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#describes
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The relation between a Description and an Entity : a Description gives a unity to a Collection of parts (the components), or constituents, by assigning a Role to each of them in the context of a whole Object (the system). A same Entity can be given different descriptions, for example, an old cradle can be given a unifying Description based on the original aesthetic design, the functionality it was built for, or a new aesthetic functionality in which it can be used as a flower pot.
Super-properties	associatedWith ^{op}
Inverse properties	is described by ^{op}
Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

directly follows ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#directlyFollows
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The intransitive follows relation. For example, Wednesday directly precedes Thursday. Directness of precedence depends on the designer conceptualization.
Super-properties	follows ^{op}
Inverse properties	directly precedes ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

directly precedes ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#directlyPrecedes
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The intransitive precedes relation. For example, Monday directly precedes Tuesday. Directness of precedence depends on the designer conceptualization.
Super-properties	precedes ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

executes task ^{op}		# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#executesTask	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A relation between an action and a task, e.g. 'putting some water in a pot and putting the pot on a fire until the water starts bubbling' executes the task 'boiling'.	
Super-properties	is classified by ^{op}	
Inverse properties	is executed in ^{op}	
Domain(s)	Action ^c	
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Task ^c	
expands ^{op}		# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#expands	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A partial order relation that holds between descriptions. It represents the proper part relation between a description and another description featuring the same properties as the former, with at least one additional one. Descriptions can be expanded either by adding other descriptions as parts, or by refining concepts that are used by them. An 'intention' to expand must be present (unless purely formal theories are considered, but even in this case a criterion of relevance is usually active).	
Super-properties	is related to description ^{op}	
Inverse properties	is expanded in ^{op}	
Domain(s)	Description ^c	
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c	
expresses ^{op}		# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#expresses	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	This is a large comment field for those who want to investigate the different uses of the 'expresses' relation for modeling different approaches to meaning characterization and modeling. For example, in all these cases, some aspect of meaning is involved: • Beehive means "a structure in which bees are kept, typically in the form of a dome or box." (Oxford dictionary) • 'Beehive' is a synonym in noun synset 09218159 "beehive hive" (WordNet) • 'the term Beehive can be interpreted as the fact of 'being a beehive', i.e. a relation that holds for concepts such as Bee, Honey, Hosting, etc.' • 'the text of Italian apiculture regulation expresses a rule by which beehives should be kept at least one kilometer away from inhabited areas' • 'the term Beehive expresses the concept Beehive' • "Beehive" for apiculturists does not express the same meaning as for, say, fishermen' • 'Your meaning of 'Beautiful' does not seem to fit mine' • "Beehive" is formally interpreted as the set of all beehives' • 'from the term 'Beehive', we can build a vector space of statistically significant cooccurring terms in the documents that contain it' • the lexeme 'Belly' expresses the role 'Body_Part' in the frame 'ObservableBodyParts' (FrameNet)	

As the examples suggest, the 'meaning of meaning' is dependent on the background approach/theory that one assumes. One can hardly make a summary of the too many approaches and theories of meaning, therefore this relation is maybe the most controversial and difficult to explain; normally, in such cases it would be better to give up formalizing. However, the usefulness of having a 'semantic abstraction' in modeling information objects is so high (e.g. for the semantic web, interoperability, reengineering, etc.), that we accept this challenging task, although without taking any particular position in the debate. We provide here some examples, which we want to generalize upon when using the 'expresses' relation to model semantic aspects of social reality.

In the most common approach, lexicographers that write dictionaries, glossaries, etc. assume that the meaning of a term is a paraphrase (or 'gloss', or 'definition'). Another approach is provided by concept schemes like thesauri and lexicons, which assume that the meaning of a term is a 'concept', encoded as a 'lemma', 'synset', or 'descriptor'. Still another approach is that of psychologists and cognitive scientists, which often assume that the meaning of an information object is a concept encoded in the mind or cognitive system of an agent. A radically different approach is taken by social scientists and semioticians, who usually assume that meanings of an information object are spread across the communication practices in which members of a community use that object. Another approach that tackles the distributed nature of meaning is assumed by geometrical models of semantics, which assume that the meaning of an InformationObject (e.g. a word) results from the set of informational contexts (e.g. within texts) in which that object is used similarly. The logical approach to meaning is still different, since it assumes that the meaning of e.g. a term is equivalent to the set of individuals that the term can be applied to; for example, the meaning of 'Ali' is e.g. an individual person called Ali, the meaning of 'Airplane' is e.g. the set of airplanes, etc. Finally, an approach taken by structuralist linguistics and frame semantics is that a meaning is the relational context in which an information object can be applied; for example, a meaning of 'Airplane' is situated e.g. in the context ('frame') of passenger airline flights.

These different approaches are not necessarily conflicting, and they mostly talk about different aspects of so-called 'semantics'. They can be summarized and modelled within DOLCE-Ultralite as follows (notice that such list is far from exhaustive):

(1) Informal meaning (as for linguistic or commonsense semantics: a distinction is assumed between (informal) meaning and reference; see *isAbout* for an alternative pattern on reference) - Paraphrase meaning (as for lexicographic semantics). Here it is modelled as the expresses relation between instances of InformationObject and different instances of InformationObject that act as 'paraphrases' - Conceptual meaning (as for 'concept scheme' semantics). Here it is modelled as the expresses relation between instances of InformationObject and instances of Concept - Relational meaning (as for frame semantics). Here it is modelled as the expresses relation between instances of InformationObject and instances of Description - Cognitive meaning (as for 'psychological' semantics). Here it is modelled as the expresses relation between any instance of InformationObject and any different instance of InformationObject that isRealizedBy a mental, cognitive or neural state (depending on which theory of mind is assumed). Such states can be considered here as instances of Process (occurring in the mind, cognitive system, or neural system of an agent) - Cultural meaning (as for 'social science' semantics). Here it is modelled as the expresses relation between instances of InformationObject and instances of SocialObject (institutions, cultural paradigms, norms, social practices, etc.) - Distributional meaning (as for geometrical models of meaning). Here it is modelled as the expresses relation between any instance of InformationObject and any different instance of InformationObject that isFormallyRepresentedIn some (geometrical) Region (e.g. a vector space)

(2) Formal meaning (as for logic and formal semantics: no distinction is assumed between informal meaning and reference, therefore between 'expresses' and 'isAbout', which can be used interchangeably) - Object-level formal meaning (as in the traditional first-order logic semantics). Here it is modelled as the expresses relation between an instance of InformationObject and an instance of Collection that isGroundingFor (in most cases) a Set; isGroundingFor is defined in the ontology: <http://www.ontologydesignpatterns.org/ont/dul/IOLite.owl> - Modal formal meaning (as in possible-world semantics). Here it is modelled as the expresses relation between an instance of InformationObject and an instance of Collection that isGroundingFor a Set, and which isPartOf some different instance of Collection that isGroundingFor a PossibleWorld

This is only a first step to provide a framework, in which one can model different aspects of meaning. A more developed ontology should approach the problem of integrating the different uses of 'expresses', so that different theories, resources, methods can interoperate.

Super-properties [associatedWith](#)^{op}

Inverse properties [is expressed by](#)^{op}

Domain(s) [Information object](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialObject>^c

[expresses concept](#)^{op}

[# OPs](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#expressesConcept>

Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an InformationObject and a Concept , e.g. the term "dog" expresses the Concept "dog". For expressing a relational meaning, see the more general object property: expresses
Super-properties	expresses ^{op}
Inverse properties	is concept expressed by ^{op}
Domain(s)	Information object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c

far from ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#farFrom
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Generic distance relation between any Entity(s). E.g. Rome is far from Beijing, astronomy is far from necromancy.
Super-properties	associatedWith ^{op}
Inverse properties	far from ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

follows ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#follows
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between entities, expressing a 'sequence' schema. E.g. 'year 2000 follows 1999', 'preparing coffee' follows 'deciding what coffee to use', 'II World War follows I World War', etc. It can be used between tasks, processes or time intervals, and subproperties would fit best in order to distinguish the different uses.
Super-properties	associatedWith ^{op}
Inverse properties	precedes ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

has common boundary ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasCommonBoundary
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl

Description	A relation to encode either formal or informal characterizations of 'boundaries' common to two different entities: an Event that ends when another begins, two abstract regions that have a common topological boundary, two objects that are said to be 'in contact' from a commonsense perspective, etc.
Super-properties	associatedWith ^{op}
Inverse properties	has common boundary ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

has component ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasComponent
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The hasProperPart relation without transitivity, holding between an Object (the system) and another (the component), and assuming a Design that structures the Object.
Super-properties	has proper part ^{op}
Inverse properties	is component of ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

has constituent ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasConstituent
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	'Constituency' depends on some layering of the world described by the ontology. For example, scientific granularities (e.g. body-organ-tissue-cell) or ontological 'strata' (e.g. social-mental-biological-physical) are typical layerings. Intuitively, a constituent is a part belonging to a lower layer. Since layering is actually a partition of the world described by the ontology, constituents are not properly classified as parts, although this kinship can be intuitive for common sense. A desirable advantage of this distinction is that we are able to talk e.g. of physical constituents of non-physical objects (e.g. systems), while this is not possible in terms of parts. Example of are the persons constituting a social system, the molecules constituting a person, the atoms constituting a river, etc. In all these examples, we notice a typical discontinuity between the constituted and the constituent object: e.g. a social system is conceptualized at a different layer from the persons that constitute it, a person is conceptualized at a different layer from the molecules that constitute them, and a river is conceptualized at a different layer from the atoms that constitute it.
Super-properties	associatedWith ^{op}
Inverse properties	is constituent of ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

has constraint ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasConstraint

Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between parameters and entities. It allows to assert generic constraints (encoded as parameters), e.g. <code>MinimumAgeForDriving</code> <code>isConstraintFor</code> John (where John is a legal subject under the TrafficLaw). The intended semantics (not expressible in OWL) is that a <code>Parameter</code> <code>isParameterFor</code> a Concept that classifies an Entity; moreover, it entails that a <code>Parameter</code> parametrizes a Region that <code>isRegionFor</code> that Entity.
Super-properties	is classified by ^{op}
Inverse properties	is constraint for ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Parameter ^c

has location ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasLocation
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A generic, relative spatial location, holding between any entities. E.g. 'the cat is on the mat', 'Omar is in Samarcanda', 'the wound is close to the femural artery'. For 'absolute' locations, see <code>SpaceRegion</code>
Super-properties	associatedWith ^{op}
Inverse properties	is location of ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

has member ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasMember
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between collections and entities, e.g. 'my collection of saxophones includes an old Adolphe Sax original alto' (i.e. my collection has member an Adolphe Sax alto).
Super-properties	associatedWith ^{op}
Inverse properties	is member of ^{op}
Domain(s)	Collection ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

has parameter ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasParameter
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl

Description	A Concept can have a Parameter that constrains the attributes that a classified Entity can have in a certain Situation, e.g. a 4WheelDriver Role definedIn the ItalianTrafficLaw has a MinimumAge parameter on the Amount 16.
Super-properties	is related to concept ^{op}
Inverse properties	is parameter for ^{op}
Domain(s)	Concept ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Parameter ^c

has part ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasPart
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A schematic relation between any entities, e.g. 'the human body has a brain as part', '20th century contains year 1923', 'World War II includes the Pearl Harbour event'. Parthood should assume the basic properties of mereology: transitivity, antisymmetry, and reflexivity (proper Parthood of course misses reflexivity). However, antisymmetry is not supported in OWL2 explicitly, therefore DUL has to adopt one of two patterns: 1) dropping asymmetry axioms, while granting reflexivity: this means that symmetry is not enforced, but permitted for the case of reflexivity. Of course, in this way we cannot prevent symmetric usages of hasPart; 2) dropping the reflexivity axiom, and enforce asymmetry: in this case, we would prevent all symmetric usages, but we loose the possibility of enforcing reflexivity, which is commonsensical in parthood. In DUL, we adopt pattern #1 for partOf, and pattern #2 for properPartOf, which seems a good approximation: due to the lack of inheritance of property characteristics, each asymmetric hasProperPart assertion would also be a reflexive hasPart assertion (reflexive reduction design pattern). Subproperties and restrictions can be used to specialize hasPart for objects, events, etc.
Super-properties	associatedWith ^{op}
Inverse properties	is part of ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

has participant ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasParticipant
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an object and a process, e.g. 'John took part in the discussion', 'a large mass of snow fell during the avalanche', or 'a cook, some sugar, flour, etc. are all present in the cooking of a cake'.
Super-properties	associatedWith ^{op}
Inverse properties	is participant in ^{op}
Domain(s)	Event ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Object ^c

has postcondition ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasPostcondition
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Direct succession applied to situations. E.g., 'A postcondition of our Plan is to have things settled'.
Super-properties	directly precedes ^{op}
Inverse properties	is postcondition of ^{op}
Domain(s)	(Event ^c or Situation ^c)
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation

has precondition ^{op}# [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasPrecondition
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Direct precedence applied to situations. E.g., 'A precondition to declare war against a foreign country is claiming to find nuclear weapons in it'.
Super-properties	directly follows ^{op}
Inverse properties	is precondition of ^{op}
Domain(s)	(Event ^c or Situation ^c)
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation

has proper part ^{op}# [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasProperPart
Description	Asymmetric (so including irreflexive) parthood.
Super-properties	has part ^{op}
Inverse properties	is proper part of ^{op}

has quality ^{op}# [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasQuality
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between entities and qualities, e.g. 'Dmitri's skin is yellowish'.
Super-properties	associatedWith ^{op}
Inverse properties	is quality of ^{op}
Domain(s)	Entity ^c

Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Quality ^c
has region ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasRegion
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between entities and regions, e.g. 'the number of wheels of that truck is 12', 'the time of the experiment is August 9th, 2004', 'the whale has been localized at 34 degrees E, 20 degrees S'.
Super-properties	associatedWith ^{op}
Inverse properties	is region for ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Region ^c
has role ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasRole
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an object and a role, e.g. the person 'John' has role 'student'.
Super-properties	is classified by ^{op}
Inverse properties	is role of ^{op}
Domain(s)	Object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Role ^c
has setting ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasSetting
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between entities and situations, e.g. 'this morning I've prepared my coffee with a new fantastic Arabica', i.e.: (an amount of) a new fantastic Arabica hasSetting the preparation of my coffee this morning.
Super-properties	associatedWith ^{op}
Inverse properties	is setting for ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation ^c
has task ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasTask

Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between roles and tasks, e.g. 'students have the duty of giving exams' (i.e. the Role 'student' hasTask the Task 'giving exams').
Super-properties	is related to concept ^{op}
Inverse properties	is task of ^{op}
Domain(s)	Role ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Task ^c

[has time interval](#)^{op} # OPs

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasTimeInterval
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The generic relation between events and time intervals.
Super-properties	has region ^{op}
Inverse properties	is time interval of ^{op}
Domain(s)	Event ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#TimeInterval ^c

[includes action](#)^{op} # OPs

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#includesAction
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between situations and actions, e.g. 'this morning I've prepared my coffee and had my fingers burnt' (i.e.: the preparation of my coffee this morning included a burning of my fingers).
Super-properties	includes event ^{op}
Inverse properties	is action included in ^{op}
Domain(s)	Situation ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Action ^c

[includes agent](#)^{op} # OPs

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#includesAgent
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between situations and persons, e.g. 'this morning I've prepared my coffee and had my fingers burnt' (i.e.: the preparation of my coffee this morning included me).
Super-properties	includes object ^{op}

Inverse properties	is agent included in ^{op}
Domain(s)	Situation ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Agent ^c
includes event ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#includesEvent
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between situations and events, e.g. 'this morning I've prepared my coffee and had my fingers burnt' (i.e.: the preparation of my coffee this morning included a burning of my fingers).
Super-properties	is setting for ^{op}
Inverse properties	is event included in ^{op}
Domain(s)	Situation ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event ^c
includes object ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#includesObject
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between situations and objects, e.g. 'this morning I've prepared my coffee and had my fingers burnt' (i.e.: the preparation of my coffee this morning included me).
Super-properties	is setting for ^{op}
Inverse properties	is object included in ^{op}
Domain(s)	Situation ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Object ^c
includesPart ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#includesPart
Super-properties	is setting for ^{op}
includes time ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#includesTime
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between situations and time intervals, e.g. 'this morning I've prepared my coffee and had my fingers burnt' (i.e.: preparing my coffee was held this morning). A data value attached to the time interval typically complements this modelling pattern.
Super-properties	is setting for ^{op}

Inverse properties	is time included in ^{op}
Domain(s)	Situation ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#TimeInterval ^c
includesWhole ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#includesWhole
Super-properties	is setting for ^{op}
introduces ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#introduces
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a Description and a SocialAgent, e.g. a Constitutional Charter introduces the SocialAgent 'PresidentOfRepublic'.
Super-properties	associatedWith ^{op}
Inverse properties	is introduced by ^{op}
Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialAgent ^c
involves agent ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#involvesAgent
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Agent participation.
Super-properties	has participant ^{op}
Inverse properties	is agent involved in ^{op}
Domain(s)	Event ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Agent ^c
is about ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isAbout
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an information object and an Entity (including information objects). It can be used to talk about entities that are references of proper nouns: the proper noun 'Leonardo da Vinci' isAbout the Person Leonardo da Vinci; as well as to talk about sets of entities that can be described by a common noun: the common noun 'person' isAbout the set of all persons in a domain of discourse, which can be represented in DOLCE-Ultralite as an individual of the class: dul:Collection. A specific sentence may use common nouns with either a singular or plural reference, or it can even refer to all possible references (e.g. in a lexicographic definition): all those uses are kinds of aboutness.

The isAbout relation is sometimes considered as reflexive, however this is semiotically inaccurate, because information can be about itself ('de dicto' usage, as in 'John is four character long'), but it is typically about something else ('de re' usage, as in 'John loves Mary'). If a reflexivity exists in general, it rather concerns its realisation, which is always associated with an event, e.g. an utterance, which makes the information denoting itself, besides its aboutness. This is implemented in DUL with the `dul:realizesSelfInformation` property, which is used with local reflexivity in the `dul:InformationRealization` class.

Super-properties [associatedWith](#)^{op}

Inverse properties [is reference of](#)^{op}

Domain(s) [Information object](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity>^c

is action included in^{op}

[# OPs](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isActionIncludedIn>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Super-properties [is event included in](#)^{op}

Domain(s) [Action](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation>^c

is agent included in^{op}

[# OPs](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isAgentIncludedIn>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Super-properties [is object included in](#)^{op}

Domain(s) [Agent](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation>^c

is agent involved in^{op}

[# OPs](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isAgentInvolvedIn>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
Agent participation.

Super-properties [is participant in](#)^{op}

Domain(s) [Agent](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event>^c

is characterized by^{op}

[# OPs](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isCharacterizedBy>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Super-properties	associatedWith ^{op}
Domain(s)	Collection ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c

is classified by ^{op}		# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isClassifiedBy	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A relation between a Concept and an Entity, e.g. 'John is considered a typical rude man'; your last concert constitutes the achievement of a lifetime; '20-year-old means she's mature enough'.	
Super-properties	associatedWith ^{op}	
Domain(s)	Entity ^c	
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c	

is component of ^{op}		# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isComponentOf	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	The asymmetric isProperPartOf relation without transitivity, holding between an Object (the system) and another (the component), and assuming a Design that structures the Object.	
Super-properties	is proper part of ^{op}	
Domain(s)	Entity ^c	
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c	

is concept expressed by ^{op}		# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isConceptExpressedBy	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	
Description	A relation between an InformationObject and a Concept , e.g. the term "dog" expresses the Concept "dog". For expressing a relational meaning, see the more general object property: expresses	
Super-properties	is expressed by ^{op}	
Domain(s)	Concept ^c	
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationObject ^c	

is concept used in ^{op}		# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isConceptUsedIn	
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl	

Description	A more generic relation holding between a Description and a Concept. In order to be used, a Concept must be previously defined in another Description
Super-properties	associatedWith ^{op}
Inverse properties	uses concept ^{op}
Domain(s)	Concept ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c

is conceptualized by^{op}# [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isConceptualizedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation stating that an Agent is internally representing a Description . E.g., 'John believes in the conspiracy theory'; 'Niels Bohr created a solar-system metaphor for his atomic theory'; 'Jacques assumes all swans are white'; 'the task force shares the attack plan'.
Super-properties	associatedWith ^{op}
Domain(s)	Social object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Agent ^c

is concretely expressed by^{op}# [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isConcretelyExpressedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an InformationRealization and a Description, e.g. 'the printout of the Italian Constitution concretelyExpresses the Italian Constitution'. It should be supplied also with a rule stating that the InformationRealization realizes an InformationObject that expresses the Description
Super-properties	associatedWith ^{op}
Domain(s)	Social object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationRealization ^c

is constituent of^{op}# [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isConstituentOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	'Constituency' depends on some layering of the world described by the ontology. For example, scientific granularities (e.g. body-organ-tissue-cell) or ontological 'strata' (e.g. social-mental-biological-physical) are typical layerings. Intuitively, a constituent is a part belonging to a lower layer. Since layering is actually a partition of the world described by the ontology, constituents are not properly classified as parts, although this kinship can be intuitive for common sense. A desirable advantage of this distinction is that we are able to talk e.g. of physical constituents of non-physical objects (e.g. systems), while this is not possible in terms of parts. Example of are the persons constituting a social system, the molecules constituting a person, the atoms constituting a river, etc. In all these examples, we notice a typical discontinuity between the constituted and the constituent object: e.g. a social system is conceptualized at a different layer from the persons that constitute it,

a person is conceptualized at a different layer from the molecules that constitute them, and a river is conceptualized at a different layer from the atoms that constitute it.

Super-properties [associatedWith](#)^{op}

Domain(s) [Entity](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity>^c

is constraint for^{op}

[# OPs](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isConstraintFor>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
A relation between parameters and entities. It allows to assert generic constraints (encoded as parameters), e.g. `MinimumAgeForDriving isConstraintFor John` (where John is a legal subject under the `TrafficLaw`). The intended semantics (not expressible in OWL) is that a `Parameter isConstraintFor` and `Entity` if the `Parameter isParameterFor` a `Concept` that classifies that `Entity`; moreover, it entails that a `Parameter` parametrizes a `Region` that `isRegionFor` that `Entity`. The use in OWL is therefore a shortcut to annotate what `Parameter` constrains what `Entity`

Super-properties [classifies](#)^{op}

Domain(s) [Parameter](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity>^c

is covered by^{op}

[# OPs](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isCoveredBy>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
A relation between concepts and collections, where a `Concept` is said to cover a `Collection`; it corresponds to a link between the (reified) intensional and extensional interpretations of a (reified) class. E.g. the collection of vintage saxophones is covered by the `Concept 'Saxophone'` with the `Parameter 'Vintage'`.

Super-properties [associatedWith](#)^{op}

Domain(s) [Collection](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept>^c

is defined in^{op}

[# OPs](#)

URI <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isDefinedIn>

Is Defined By <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl>

Description
A relation between a `Description` and a `Concept`, e.g. a `Workflow` for a governmental `Organization` defines the Role 'officer', or 'the Italian Traffic Law defines the role Vehicle'.

Super-properties [is concept used in](#)^{op}

Domain(s) [Concept](#)^c

Range(s) <http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description>^c

is described by^{op} # OPs

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isDescribedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The relation between an Entity and a Description: a Description gives a unity to a Collection of parts (the components), or constituents, by assigning a Role to each of them in the context of a whole Object (the system). A same Entity can be given different descriptions, for example, an old cradle can be given a unifying Description based on the original aesthetic design, the functionality it was built for, or a new aesthetic functionality in which it can be used as a flower pot.
Super-properties	associatedWith ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c

is event included in^{op} # OPs

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isEventIncludedIn
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-properties	has setting ^{op}
Domain(s)	Event ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation ^c

is executed in^{op} # OPs

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isExecutedIn
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an action and a task, e.g. 'putting some water in a pot and putting the pot on a fire until the water starts bubbling' executes the task 'boiling'.
Super-properties	classifies ^{op}
Domain(s)	Task ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Action ^c

is expanded in^{op} # OPs

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isExpandedIn
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A partial order relation that holds between descriptions. It represents the proper part relation between a description and another description featuring the same properties as the former, with at least one additional one. Descriptions can be expanded either by adding other descriptions as parts, or by refining concepts that are used by them. An 'intention' to expand must be present (unless purely formal theories are considered, but even in this case a criterion of relevance is usually active).
Super-properties	is related to description ^{op}

Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c
is expressed by ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isExpressedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a <code>dul:SocialObject</code> (the 'meaning') and a <code>dul:InformationObject</code> (the 'expression'). For example: 'A Beehive is a structure in which bees are kept, typically in the form of a dome or box.' (Oxford dictionary); 'the term Beehive expresses the concept Beehive in my apiculture ontology'. The intuition for 'meaning' is intended to be very broad. A separate, large comment is included in the encoding of 'expresses', for those who want to investigate more on what kind of meaning can be represented in what form.
Super-properties	associatedWith ^{op}
Domain(s)	Social object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationObject ^c
is introduced by ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isIntroducedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a <code>Description</code> and a <code>SocialAgent</code> , e.g. a Constitutional Charter introduces the <code>SocialAgent</code> 'PresidentOfRepublic'.
Super-properties	associatedWith ^{op}
Domain(s)	Social agent ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c
is location of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isLocationOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A generic, relative localization, holding between any entities. E.g. 'Rome is the seat of the Pope', 'the liver is the location of the tumor'. For 'absolute' locations, see <code>SpaceRegion</code>
Super-properties	associatedWith ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c
is member of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isMemberOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl

Description	A relation between collections and entities, e.g. 'the Night Watch by Rembrandt is in the Rijksmuseum collection'; 'Davide is member of the Pen Club', 'Igor is one the subjects chosen for the experiment'.
Super-properties	associatedWith ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Collection ^c

is object included in ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isObjectIncludedIn
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-properties	has setting ^{op}
Domain(s)	Object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation ^c

is observable at ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isObservableAt
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation to represent a (past, present or future) TimeInterval at which an Entity is observable. In order to encode a specific time, a data value should be related to the TimeInterval. An alternative way of representing time is the datatype property: hasIntervalDate
Super-properties	has region ^{op}
Inverse properties	is time of observation of ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#TimeInterval ^c

is parameter for ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isParameterFor
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Concept can have a Parameter that constrains the attributes that a classified Entity can have in a certain Situation, e.g. a 4WheelDriver Role definedIn the ItalianTrafficLaw has a MinimumAge parameter on the Amount 16.
Super-properties	is related to concept ^{op}
Domain(s)	Parameter ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c

is parametrized by ^{op} # OPs	
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isParametrizedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The relation between a Parameter, e.g. 'MajorAge', and a Region, e.g. '>17 year'.
Super-properties	is classified by ^{op}
Inverse properties	parametrizes ^{op}
Domain(s)	Region ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Parameter ^c

is part of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isPartOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between any entities, e.g. 'brain is a part of the human body'. See dul:hasPart for additional documentation.
Super-properties	associatedWith ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

is participant in ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isParticipantIn
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an object and a process, e.g. 'John took part in the discussion', 'a large mass of snow fell during the avalanche', or 'a cook, some sugar, flour, etc. are all present in the cooking of a cake'.
Super-properties	associatedWith ^{op}
Domain(s)	Object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event ^c

is postcondition of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isPostconditionOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Direct succession applied to situations. E.g., 'Taking some rest is a postcondition of my search for a hotel'.
Super-properties	directly follows ^{op}
Domain(s)	(Event ^c or Situation ^c)

Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation
is precondition of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isPreconditionOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Direct precedence applied to situations. E.g., 'claiming to find nuclear weapons in a foreign country is a precondition to declare war against it'.
Super-properties	directly precedes ^{op}
Domain(s)	(Event ^c or Situation ^c)
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation
is proper part of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isProperPartOf
Description	See dul:hasProperPart for additional documentation.
Super-properties	is part of ^{op}
is quality of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isQualityOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between entities and qualities, e.g. 'Dmitri's skin is yellowish'.
Super-properties	associatedWith ^{op}
Domain(s)	Quality ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c
is realized by ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isRealizedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an information realization and an information object, e.g. the paper copy of the Italian Constitution realizes the text of the Constitution.
Super-properties	associatedWith ^{op}
Inverse properties	realizes ^{op}
Domain(s)	Information object ^c

Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationRealization ^c
is reference of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isReferenceOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between information objects and any Entity (including information objects). It can be used to talk about e.g. entities are references of proper nouns: the proper noun 'Leonardo da Vinci' isAbout the Person Leonardo da Vinci; as well as to talk about sets of entities that can be described by a common noun: the common noun 'person' isAbout the set of all persons in a domain of discourse, which can be represented in DOLCE-Ultralite as an individual of the class: Collection . The isReferenceOf relation is irreflexive, differently from its inverse isAbout.
Super-properties	associatedWith ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationObject ^c
is reference of information realized by ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isReferenceOfInformationRealizedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The relation between entities and information realizations, e.g. between Italy and a paper copy of the text of the Italian Constitution.
Super-properties	associatedWith ^{op}
Inverse properties	realizes information about ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationRealization ^c
is region for ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isRegionFor
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between entities and regions, e.g. 'the color of my car is red'.
Super-properties	associatedWith ^{op}
Domain(s)	Region ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c
is related to concept ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isRelatedToConcept

Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any relation between concepts, e.g. superordinated, conceptual parthood, having a parameter, having a task, superordination, etc.
Super-properties	associatedWith ^{op}
Inverse properties	is related to concept ^{op}
Domain(s)	Concept ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c

[is related to description](#)^{op} # [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isRelatedToDescription
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Any relation between descriptions.
Super-properties	associatedWith ^{op}
Inverse properties	is related to description ^{op}
Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c

[is role defined in](#)^{op} # [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isRoleDefinedIn
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a description and a role, e.g. the role 'Ingredient' is defined in the recipe for a cake.
Super-properties	is defined in ^{op}
Domain(s)	Role ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c

[is role of](#)^{op} # [OPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isRoleOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an object and a role, e.g. 'student' is the role of 'John'.
Super-properties	classifies ^{op}
Domain(s)	Role ^c

Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Object ^c
is satisfied by ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isSatisfiedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a Situation and a Description, e.g. the execution of a Plan satisfies that plan.
Super-properties	associatedWith ^{op}
Inverse properties	satisfies ^{op}
Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation ^c
is setting for ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isSettingFor
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between situations and entities, e.g. 'this morning I've prepared my coffee with a new fantastic Arabica', i.e.: the preparation of my coffee this morning is the setting for (an amount of) a new fantastic Arabica.
Super-properties	associatedWith ^{op}
Domain(s)	Situation ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c
is specialized by ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isSpecializedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A partial order relation that holds between social objects. It represents the subsumption relation between e.g. a Concept and another Concept that is broader in extensional interpretation, but narrower in intensional interpretation. E.g. PhDStudent Role specializes Student Role
Super-properties	associatedWith ^{op}
Inverse properties	specializes ^{op}
Domain(s)	Social object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialObject ^c
is subordinated to ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isSubordinatedTo
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl

Description	Direct succession applied to concepts. E.g. the role 'Officer' is subordinated to 'Director'.
Super-properties	is related to concept ^{op} directly follows ^{op}
Inverse properties	is superordinated to ^{op}
Domain(s)	Concept ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c
is superordinated to ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isSuperordinatedTo
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Direct precedence applied to concepts. E.g. the role 'Executive' is superordinated to 'DepartmentManager'.
Super-properties	is related to concept ^{op} directly precedes ^{op}
Domain(s)	Concept ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c
is task defined in ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isTaskDefinedIn
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a description and a task, e.g. the task 'boil' is defined in a recipe for a cake.
Super-properties	is defined in ^{op}
Domain(s)	Task ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c
is task of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isTaskOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between roles and tasks, e.g. 'students have the duty of giving exams' (i.e. the Role 'student' hasTask the Task 'giving exams').
Super-properties	is related to concept ^{op}
Domain(s)	Task ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Role ^c
is time included in ^{op} # OPs	

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isTimeIncludedIn
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-properties	has setting ^{op}
Domain(s)	Time interval ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Situation ^c

is time interval of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isTimeIntervalOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The generic relation between time intervals and events.
Super-properties	is region for ^{op}
Domain(s)	Time interval ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Event ^c

is time of observation of ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isTimeOfObservationOf
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation to represent a (past, present or future) TimeInterval at which an Entity is observable. In order to encode a specific time, a data value should be related to the TimeInterval. An alternative way of representing time is the datatype property: hasIntervalDate
Super-properties	is region for ^{op}
Domain(s)	Time interval ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

is unified by ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#isUnifiedBy
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Collection has a unification criterion, provided by a Description; for example, a community of practice can be unified by a shared theory or interest, e.g. the community that makes research on mirror neurons shares some core knowledge about mirror neurons, which can be represented as a Description MirrorNeuronTheory that unifies the community. There can be several unifying descriptions.
Super-properties	associatedWith ^{op}
Inverse properties	unifies ^{op}
Domain(s)	Collection ^c

Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c
near to ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#nearTo
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Generic distance relation between any Entity(s). E.g. Rome is near to Florence, astronomy is near to physics.
Super-properties	associatedWith ^{op}
Inverse properties	near to ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c
overlaps ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#overlaps
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A schematic relation between any entities, e.g. 'the chest region overlaps with the abdomen region', 'my spoken words overlap with hers', 'the time of my leave overlaps with the time of your arrival', 'fibromyalgia overlaps with other conditions'. Subproperties and restrictions can be used to specialize overlaps for objects, events, time intervals, etc.
Super-properties	associatedWith ^{op}
Inverse properties	overlaps ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c
parametrizes ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#parametrizes
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The relation between a Parameter, e.g. 'MajorAgeLimit', and a Region, e.g. '18_year'. For a more data-oriented relation, see hasDataValue
Super-properties	classifies ^{op}
Domain(s)	Parameter ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Region ^c
precedes ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#precedes

Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between entities, expressing a 'sequence' schema. E.g. 'year 1999 precedes 2000', 'deciding what coffee to use' precedes 'preparing coffee', 'World War II follows World War I', 'in the Milan to Rome autoroute, Bologna precedes Florence', etc. It can then be used between tasks, processes, time intervals, spatially locate objects, situations, etc. Subproperties can be defined in order to distinguish the different uses.
Super-properties	associatedWith ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

realizes ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#realizes
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between an information realization and an information object, e.g. the paper copy of the Italian Constitution realizes the text of the Constitution.
Super-properties	associatedWith ^{op}
Domain(s)	Information realization ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#InformationObject ^c

realizes information about ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#realizesInformationAbout
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	The relation between entities and information realizations, e.g. between Italy and a paper copy of the text of the Italian Constitution.
Super-properties	associatedWith ^{op}
Domain(s)	Information realization ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

realizesSelfInformation ^{op}	# OPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#realizesSelfInformation
Description	This relation is a workaround to enable local reflexivity axioms (Self) working with non-simple properties; in this case, dul:realizesInformation About.
Super-properties	realizes information about ^{op}

is in the same setting as ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#sameSettingAs
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Super-properties	associatedWith ^{op}
Inverse properties	is in the same setting as ^{op}
Domain(s)	Entity ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Entity ^c

satisfies ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#satisfies
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A relation between a Situation and a Description, e.g. the execution of a Plan satisfies that plan.
Super-properties	associatedWith ^{op}
Domain(s)	Situation ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Description ^c

specializes ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#specializes
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A partial order relation that holds between social objects. It mainly represents the subsumption relation between e.g. a Concept or Description and another Concept (resp. Description) that is broader in extensional interpretation, but narrower in intensional interpretation. For example, the role PhDStudent specializes the role Student. Another possible use is between a Collection that isCoveredBy a Concept A, and another Collection that isCoveredBy a Concept B that on its turn specializes A. For example, the 70,000 series Selmer Mark VI saxophone Collection specializes the Selmer Mark VI saxophone Collection.
Super-properties	associatedWith ^{op}
Domain(s)	Social object ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#SocialObject ^c

unifies ^{op}	# OPs
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URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#unifies
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A Collection has a unification criterion, provided by a Description; for example, a community of practice can be unified by a shared theory or interest, e.g. the community that makes research on mirror neurons shares some core knowledge about mirror neurons, which can be represented as a Description MirrorNeuronTheory that unifies the community. There can be several unifying descriptions.
Super-properties	associatedWith ^{op}

Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Collection ^c
uses concept ^{op} # OPs	
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#usesConcept
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A generic relation holding between a Description and a Concept. In order to be used, a Concept must be previously defined in another Description. This last condition cannot be encoded for object properties in OWL.
Super-properties	associatedWith ^{op}
Domain(s)	Description ^c
Range(s)	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#Concept ^c

Datatype Properties



has data value has event date has interval date has parameter data value has region data value	
has data value ^{dp}	# DPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasDataValue
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A datatype property that encodes values from a datatype for an Entity. There are several ways to encode values in DOLCE (Ultralite): 1) Directly assert an xsd: value to an Entity by using <i>hasDataValue</i> 2) Assert a Region for an Entity by using <i>hasRegion</i>, and then assert an xsd: value to that Region, by using <i>hasRegionDataValue</i> 3) Assert a Quality for an Entity by using <i>hasQuality</i>, then assert a Region for that Quality, and assert an xsd: value to that Region, by using <i>hasRegionDataValue</i> 4) When the value is required, but not directly observed, assert a Parameter for an xsd: value by using <i>hasParameterDataValue</i>, and then associate the Parameter to an Entity by using <i>isConstraintFor</i> 5) When the value is required, but not directly observed, you can also assert a Parameter for a Region by using <i>parametrizes</i>, and then assert an xsd:_ value to that Region, by using <i>hasRegionDataValue</i> The five approaches obey different requirements. For example, a simple value can be easily asserted by using pattern (1), but if one needs to assert an interval between two values, a Region should be introduced to materialize that interval, as pattern (2) suggests. Furthermore, if one needs to distinguish the individual Quality of a value, e.g. the particular nature of the density of a substance, pattern (3) can be used. Patterns (4) and (5) should be used instead when a constraint or a selection is modeled, independently from the actual observation of values in the real world.
Domain(s)	Entity ^c
has event date ^{dp}	# DPs
URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasEventDate
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A datatype property that encodes values from xsd:dateTime for an Event; a same Event can have more than one xsd:dateTime value: begin date, end date, date at which the interval holds, etc.

Super-properties	has data value ^{dp}
Domain(s)	Event ^c
Range(s)	xsd:dateTime ^c

[has interval date](#) ^{dp} [# DPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasIntervalDate
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A datatype property that encodes values from xsd:dateTime for a TimeInterval; a same TimeInterval can have more than one xsd:dateTime value: begin date, end date, date at which the interval holds, etc.
Super-properties	has region data value ^{dp}
Domain(s)	Time interval ^c
Range(s)	xsd:dateTime ^c

[has parameter data value](#) ^{dp} [# DPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasParameterDataValue
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	Parametrizes values from a datatype. For example, a Parameter MinimumAgeForDriving hasParameterDataValue 18 on datatype xsd:int, in the Italian traffic code. In this example, MinimumAgeForDriving isDefinedIn the Norm ItalianTrafficCodeAgeDriving. More complex parametrization requires workarounds. E.g. AgeRangeForDrugUsage could parametrize data value: 14 to 50 on the datatype: xsd:int. Since complex datatypes are not allowed in OWL1.0, a solution to this can only work by creating two 'sub-parameters': MinimumAgeForDrugUsage (that hasParameterDataValue 14) and MaximumAgeForDrugUsage (that hasParameterDataValue 50), which are components of (cf. hasComponent) the main Parameter AgeRangeForDrugUsage. Ordering on subparameters can be created by using or specializing the object property 'precedes'.
Super-properties	has data value ^{dp}
Domain(s)	Parameter ^c

[has region data value](#) ^{dp} [# DPs](#)

URI	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl#hasRegionDataValue
Is Defined By	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
Description	A datatype property that encodes values for a Region, e.g. a float for the Region Height.
Super-properties	has data value ^{dp}
Domain(s)	Region ^c

Annotation Properties



[creator](#) [date](#) [comment](#) [label](#) [versionInfo](#)

creator ^{ap}	# APs
URI	http://purl.org/dc/elements/1.1/creator
date ^{ap}	# APs
URI	http://purl.org/dc/elements/1.1/date
comment ^{ap}	# APs
URI	http://www.w3.org/2000/01/rdf-schema#comment
label ^{ap}	# APs
URI	http://www.w3.org/2000/01/rdf-schema#label
versionInfo ^{ap}	# APs
URI	http://www.w3.org/2002/07/owl#versionInfo

Namespaces



default (:)

	http://www.ontologydesignpatterns.org/ont/dul/DUL.owl
dc	http://purl.org/dc/elements/1.1/
dct	http://purl.org/dc/terms/
owl	http://www.w3.org/2002/07/owl#
prov	http://www.w3.org/ns/prov#
rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs	http://www.w3.org/2000/01/rdf-schema#
sdo	https://schema.org/
skos	http://www.w3.org/2004/02/skos/core#
xsd	http://www.w3.org/2001/XMLSchema#

Legend

- c Classes
- op Object Properties
- fp Functional Properties
- dp Data Properties

- dp** Annotation Properties
- p** Properties
- ni** Named Individuals